

MECHANICS OF COMPOSITE STRUCTURES

VII Semester

Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6AE38		PCC		L	T	P	C	CIA	SEE	Total
				3	0	0	3	40	60	100
UNIT-I		CHARACTERISTICS AND APPLICATIONS OF COMPOSITES								
Introduction to materials, natural and man-made composites, Aerospace and structural applications, types and classification of composites. Reinforcements: Fibres – Glass, Silica, Kevlar, carbon, boron, silicon carbide, and boron carbide fibres. Particulate composites, Polymer composites, Thermoplastics, Thermosets, Metal matrix and ceramic composites.										
UNIT-II		MICROMECHANICS								
Unidirectional composites, constituent materials and properties, elastic properties of a lamina, properties of typical composite materials, laminate characteristics and configurations. Characterization of composite properties. Advanced manufacturing methods										
UNIT-III		COORDINATE TRANSFORMATION								
Hooke's law for different types of materials, Hooke's law for two dimensional unidirectional lamina, Transformation of stress and strain, Numerical examples of stress strain transformation										
Elastic behavior of unidirectional composites: Elastic constants of lamina, relationship between engineering constants and reduced stiffness and compliances, analysis of laminated composites, constitutive relations.										
UNIT-IV		STRENGTH OF UNIDIRECTIONAL LAMINA								
Micro mechanics of failure, Failure mechanisms, strength of an orthotropic lamina, strength of a lamina under tension and shear maximum stress and strain criteria, application to design. The failure envelope, first ply failure, free-edge effects. Micro mechanical predictions of elastic constants.										
UNIT-V		ANALYSIS OF LAMINATED COMPOSITE PLATES								
Introduction thin plate theory, specially orthotropic plate, cross and angle ply laminated plates, problems using thin plate theory.										
Text Books:										
1. R. M. Jones, "Mechanics of Composite Materials", 2nd Edition, Taylor & Francis 2. Madhujit Mukhopadhyay, "Mechanics of Composite Materials and Structures", Orient Blackswan Pvt Ltd.										
Reference Books:										
1.D. Agarwal, L. J. Broutman and K. Chandrashekhara, "Analysis and Performance of Fiber Composites", 3rd Edition, John Wiley & Sons. 2. R.F. Gibson, "Principles of composite material mechanics", 3rd Edition, CRC press.										
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:										
1. Exemplify the properties and applications of composite materials used in aircraft 2. Determine the material properties of the composites using micromechanics approach 3. Exemplify the transformation of stress and strain in composite material 4. Illustrate the elastic behaviour of unidirectional composites subjected to stress 5. Predict the strength of the unidirectional lamina and laminate by major analysing techniques										